Exc. no: 2 Study of different types of Network cables

Aim:- Study of different types of Network cables

a). Understand different types of network cable:-

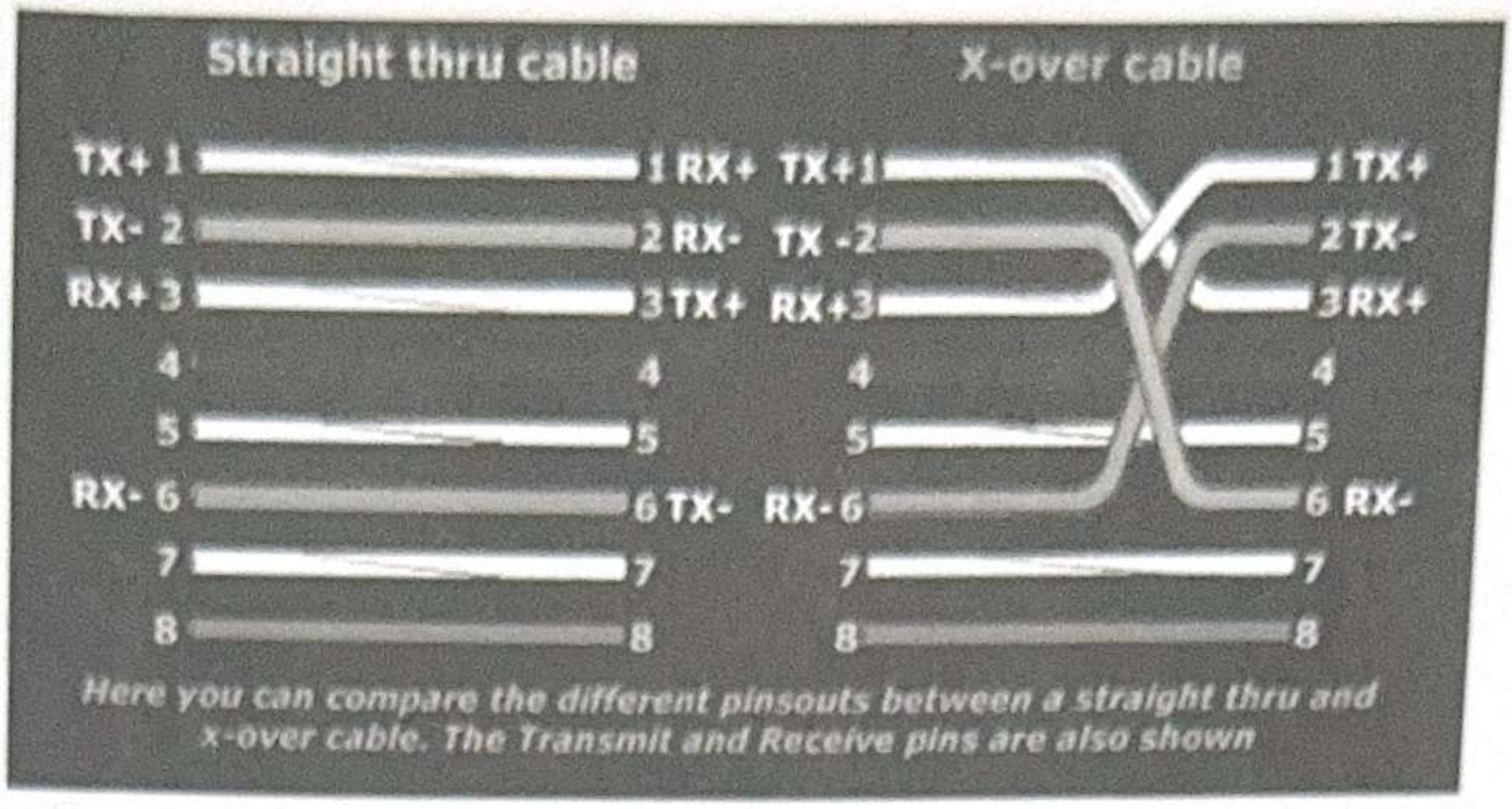
1. Unshielded Twisted Pair (UTP) Cable.
2. Shielded Twisted Pair (STP) Cable.
3. Coaxial Cable.
4. Fibre Optic Cable.

Cable Type	Category	Max Data Transmission	Advantages & Disadvantage	Application / Use
UTP	Category 3	10 Mbps	Advantages:- • Cheap • Easy to install Disadvantage:- • Not good to EMI	10 Base-T, Fast, Gigabyte Ethernet
	Category 5	upto 100 Mbps		
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages:- • Shielded • Faster than UTP • Less Susceptible Disadvantage:- • Expensive	Gigabyte, 10G Ethernet (55m) Gigabyte, 10G (100m) Ethernet
SSTP	Category 7	10 Gbps		
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages:- • High bandwidth • Immune to interference • Versatile Disadvantage:- • Limited distance • Cost • Size is bulky	Speed of signal is 500m TV network High speed internet Connection
Fibre optics Cable	Single mode Multi mode	100 Gbps	Advantages:- • High speed • High bandwidth • High security • Long distance Disadvantages:- • Expensive • Required skilled installers	Maximum distance of fibre optics cable is around 100 meters

b). Make your own Ethernet Cross over Cable / Straight Cable:-

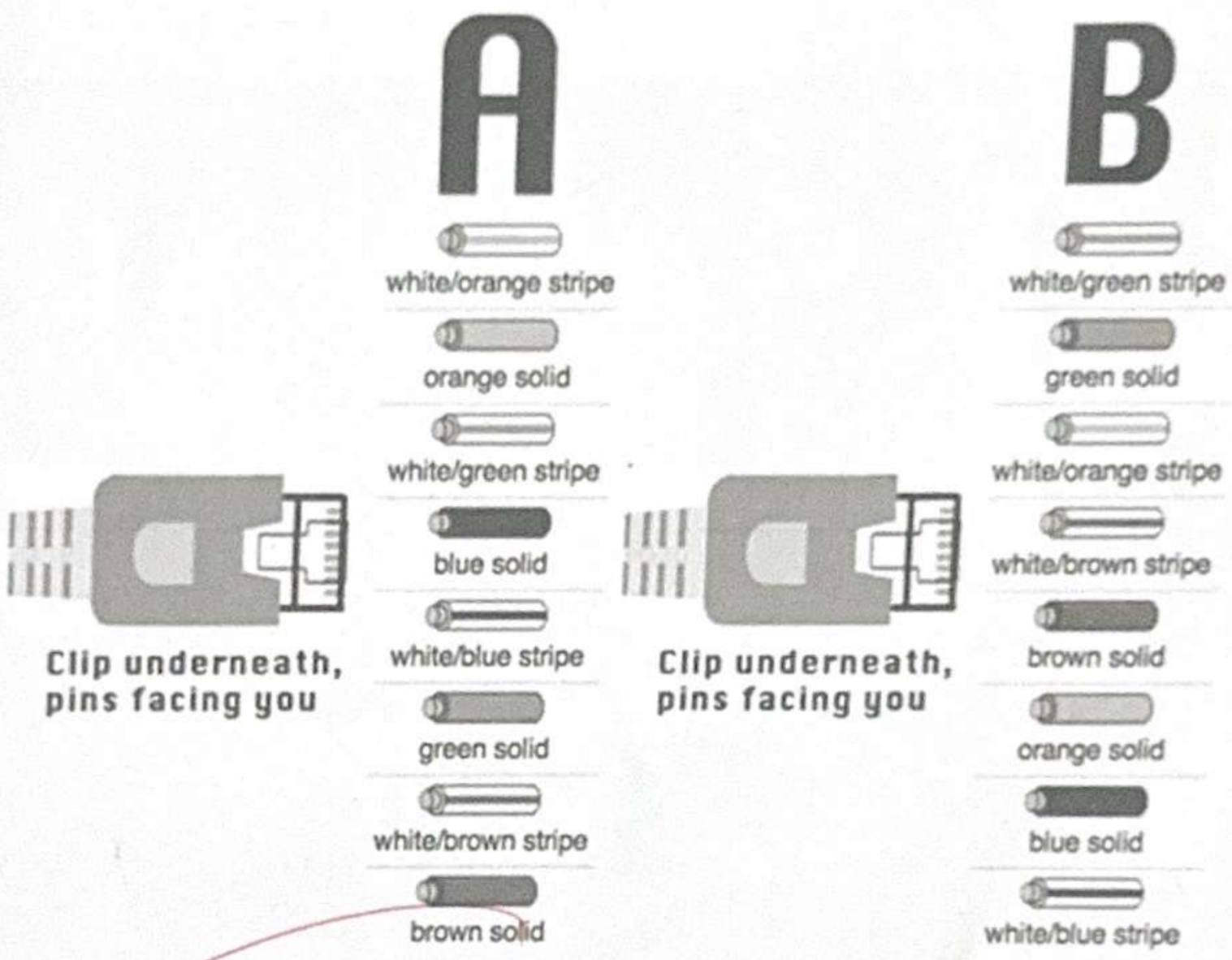
Tools & parts needed:

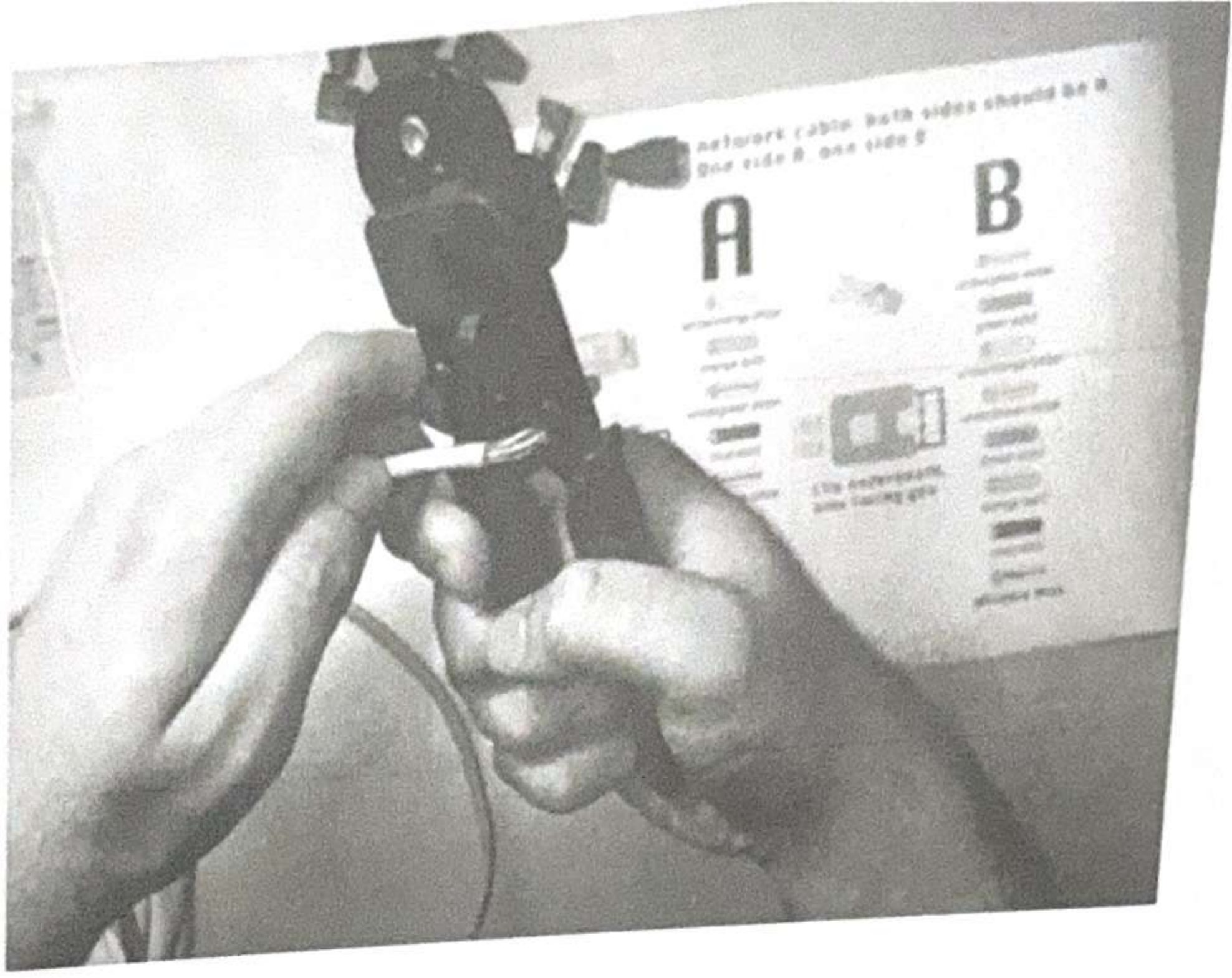
- Ethernet Cable - CAT5e is certified but CAT5 also works.
- A crimping tool - All-in-one networking tool
- Two RJ45 plugs
- Optional two plug, shields.



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Straight through network cable: both sides should be A
Crossover cable: One side A, one side B




network cable. both sides should be a
one side a. one side b
A
B


Cros
A
underneath,
facing you

Step 1:- To start construction of the device, begin by threading shields onto the cable

Step 2:- Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task -

Step 3:- After, you will need to charge the wires, there should be four "twisted pairs". One end should be in arrangement A & the other is B

Step 4:- Once the order is correct, bunch the together in a line. To do so, hold the plug with the clip side facing away from you & have the gold pins facing toward you, as shown.

Step 5:- Next push the cable right in. The notch at the end of the plug needs to be just over the cable shielding & if it isn't, that means that you stripped off too much shielding.

Step 6:- After the wires are securely sitting inside the plug, insert it into the crimping tool & push down

Step 7:- Lastly, repeat the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable)

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Student Observation:-

1). What is the difference between cross cable & straight cable?

Ans: Straight Cable: Same pin standards on both ends; connects difference device type

Cross Cable: Swapped pin standards on ends; connects identical device types.

2). Which type of cable is used to connect two PC?

Ans Cross cable.

3). Which type of cable is used to connect a router/ switch to your PC?

Ans:- Switch to PC: Straight Cable

Router to PC: Cross Cable

4). Find out category of twisted pair cable used in your lab?

Ans CAT6

5). Write down your understanding, challenges faced & output received while making a twisted pair (cross) cable.

Ans Understanding: Wires must match specific color standards to transmit data.

Challenges:- Lining up the tiny wires & keeping them in order inside the RJ45 plug.

Output:- A working cable that passing a LAN tester checker.